Delay Trio Manual

Understanding the three effects

Tom Erbe developed the original trio while teaching computer music and studio technique at UC San Diego. They share a delay-line approach: sound is recorded into a buffer, read back later, and optionally returned through a feedback loop. Smooth interpolation helps read between stored samples when the playback position moves.

Delayed presents a conventional delay with a resonant feedback filter and an LFO that moves the read position. **Pitch Delay** uses two moving playback heads and crossfades between them to shift pitch. **Bubbler** reads short, enveloped fragments from the buffer and overlaps them into a granular texture. These are three related ways of reading and recombining the recorded sound.

Time sets the underlying delay duration. Its logarithmic response gives more encoder travel to short times. Very short times can produce comb-filter, flanging or metallic colors; longer times separate the repeats. Moving a read position can change pitch as well as delay time.

Feedback returns the delayed signal to the buffer. Low amounts give a few repeats; higher amounts extend the tail. Delayed and Pitch Delay reach 200%, allowing the returning signal to exceed unity gain. Their feedback paths use saturation and DC blocking to control excessive levels and offsets, but high feedback can still build into strong, distorted or sustained sound. Bubbler uses a reduced 0–95% range.

Mix blends the original and processed signals: 0% is dry, 100% is wet. For an obvious first listen, raise Mix and begin with moderate Feedback. At 100% wet, the direct attack is replaced by the delayed or grain-based output.

Bubbler • Delayed • Pitch Delay

Control images throughout this manual identify the encoders and buttons on the supplied OP-1 illustration. “Push” means pressing an encoder; “hold” means keeping the button pressed while operating another control.


Blue encoder


Green encoder


White encoder


Orange encoder

The devices are OP-1 adaptations of SoundHack Delay Trio by Tom Erbe, built with the Opus framework. Ports and interface adaptation by ESC_CODE.

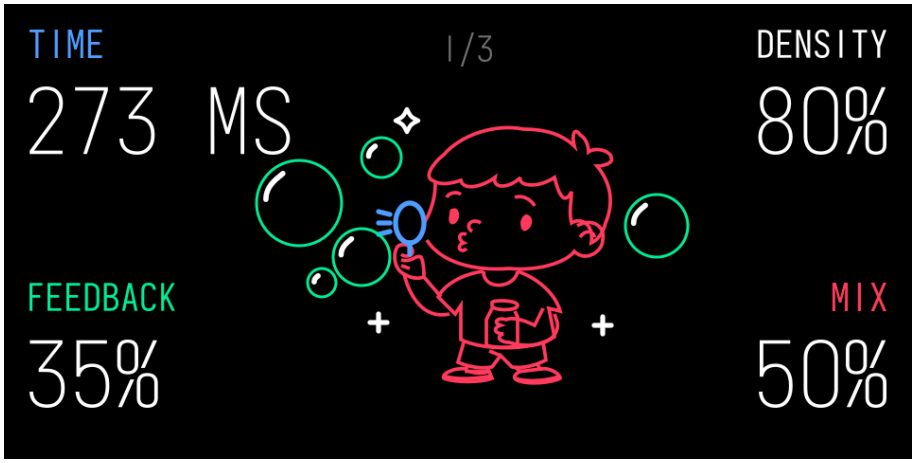
OP-1 adaptation of **SoundHack Delay Trio by Tom Erbe**. Expanded using the original SoundHack manual (2008), with descriptions checked against the ESC_CODE ports. The controls and ranges below describe the OP-1 devices.

Control reference: Bubbler 1.1.12, Delayed 0.3.4 and Pitch Delay 0.3.3. The finer Density encoder response introduced in the Bubbler 1.1.13 test build is noted separately.

Installation and common behavior

For the original OP-1, firmware 246, with Opus 2.6. Copy bubbler.opmod, delayed.opmod and pitchdelay.opmod to mods, safely eject and restart. Choose the effect on a synth or drum instrument. All three are instrument-only; selecting them as master effects shows **instrument only**.

Bubbler



A granular delay: overlapping, smoothly enveloped pieces of recorded sound are recombined into the output and feedback.

Delay time, grain starts, size, direction and octave can vary independently.

Time, Feedback and Density

Page 1	Control
Blue	Time: 30 ms–1 second
Green	Feedback: 0–95%
White	Density: 0–100%, controlling grain activity
Orange	Mix: dry/wet balance

Time sets the base delay from which grains read. **Feedback** sends the recombined grain output back into the delay, so repeated material can be fragmented and transformed again on later passes.

Density controls grain activity. At 0%, no new grains are scheduled; grains already playing can finish.

Increasing it creates more overlap and a more continuous cloud. The OP-1 engine supports up to **eight simultaneous grains**. Size, start variation and available grain slots affect the actual overlap.

Page 2 & 3

Hold Shift **Shift** to access the additional pages. While holding Shift Shift, > Right goes forward and < Left goes backward between pages 2 and 3. Release Shift Shift to return to page 1. The next Shift Shift press recalls the last additional page. The page counter shows your position.

Grain timing and size

Page 2	Control
Blue	Time Var: random variation of each grain’s delay time
Green	Start Var: variation of grain start timing
White	Size: grain duration, 2–50% of Time
Orange	Size Var: random variation of grain duration

Time Var chooses a different read delay for each new grain. At zero, grains use the common base delay. Increasing it spreads their read positions, making echoes less regular.

Start Var varies when grains actually begin. With zero variation, scheduling is regular and depends on Size and Density. Raising it loosens that rhythm.

This differs from Time Var: Start Var changes the timing of grain entrances, while Time Var changes which part of the recorded sound each grain reads.

Size sets a grain's duration as 2–50% of the current Time. For example, at Time 400 ms and Size 25%, the nominal grain is 100 ms long.
Small grains can sound buzzy or fragmented; larger grains preserve longer recognizable pieces. Each grain has a smooth amplitude envelope to soften its boundaries. Overlap helps join those envelopes into continuous sound.

Size Var gives grains different durations around the nominal Size. At zero, their lengths match; higher values make the texture less uniform. Minimum grain length and buffer/pitch limits can constrain the result.

Grain pitch, direction and tone

Page 3	Control
Blue	Pitch: -12, -7, -5, -3, Off, key, +3, +5, +7, +12 semitones
Green	Pitch Var: probability of adding a random octave shift to each grain
White	Filter Freq: feedback filter cutoff
White push	Toggle the white encoder between Filter Freq and Reso
Orange	Reverse: probability that a grain plays backward

Pitch supplies the base transposition for newly created grains.

- Off means unison with no keyboard-controlled transposition; it does not disable the effect.
- KEY follows the notes on the keyboard.
- The interval choices are equal-tempered semitone shifts.

A grain keeps the pitch chosen when it starts, so a control change can overlap old-pitch and new-pitch grains briefly.

At **Pitch Var** 0%, no random octave is added; at 100%, every grain receives a random -12 or +12 semitone shift. This is added to the selected base pitch.

Reverse is a probability per new grain: 0% plays all grains forward; 100% plays all grains backward. Intermediate values mix the two directions. It is especially easy to hear with longer grains and more recognizable source material.

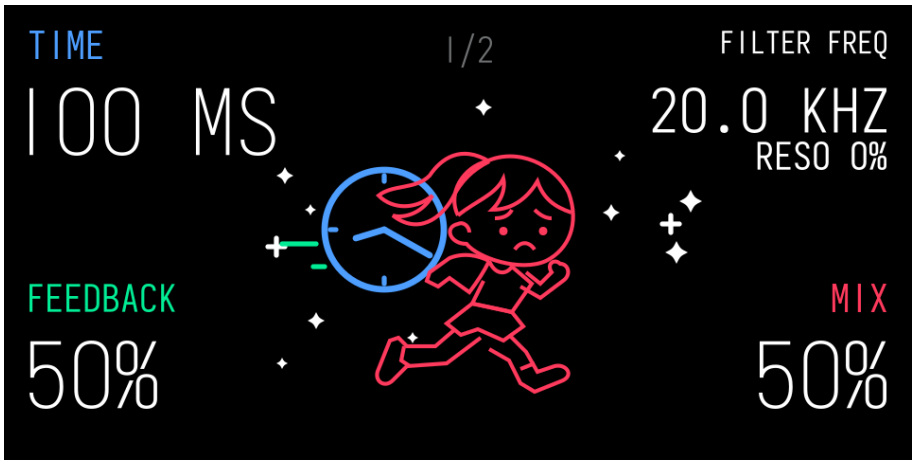
Filter Freq sets the feedback filter cutoff; lower values reduce high-frequency content in subsequent repeats.

Reso emphasizes sound near that cutoff. Higher resonance can ring and change the character or level of the recirculating material.

Try it

Start with a low Density and a large Size to hear individual fragments. Increase Density to join them into a cloud, then add Start Var and Time Var to loosen the pattern. For backward swells, increase Reverse and Size. For shifting layers, choose a base Pitch and introduce Pitch Var gradually.

Delayed



A delay with a resonant lowpass filter in its feedback loop and LFO modulation of the read delay.

Moving the read position produces vibrato, chorus-like motion and more extreme time/pitch effects.

The sine LFO uses interpolation, and the feedback filter retains extra precision for clean decay tails.

Main page	Control
Blue	Time: 1 ms–1 second
Green	Feedback: 0–200%
White	Filter Freq: 20 Hz–20 kHz
Orange	Mix

White push switches between Filter Freq and Reso.

Orange push toggles Freeze. It stops admitting new input and recirculates a captured loop of the current delay content. Press again to resume normal input. Ice and white artwork indicate Freeze.

Filter and feedback

Filter Freq spans 20 Hz–20 kHz with an exponential response. It is in the feedback path, so filtering affects material written back into the delay.

At 20 kHz the filter is bypassed. Lowering the cutoff progressively darkens the repeats.

Reso emphasizes the region around the cutoff. The original manual describes settings above about 50% as strongly resonant; use that as a listening guide rather than a guarantee of identical response in this fixed-point port. High resonance can ring, especially with strong Feedback.

Feedback above 100% can build up strongly; raise it gradually. At short delay times, the returned signal interacts with the input as a comb-filter-like texture.

Changing Time while sound is circulating can make pitch sweeps.

Modulation

Hold Shift	Control
Blue	LFO Freq: 0.01–10 Hz
Green	LFO Depth: 0–100%
White	LFO Shape: Sine, Tri, Square, Up, Down, Rand
Orange	LFO Phase: -180° to +180°

How modulation works

LFO Freq is the speed of the repeating modulation. The exponential range spans 0.01 Hz (one cycle every 100 seconds) to 10 Hz (ten cycles per second).

LFO Depth sets how far the delay moves relative to Time. With a symmetric waveform, a nominal Time of 400 ms and Depth of 25% sweeps approximately from 300 to 500 ms. The engine clamps read positions to its available history and minimum safe delay. The visible Time setting remains capped at one second; modulation can move the internal read delay beyond that base setting within the buffer's limits.

LFO Shape changes the motion:

Shape	Behavior
Sine	Smooth curved motion; useful for gentle modulation
Tri	A steady rise and fall
Square	Switches abruptly between two delay positions; can produce intentional clicks or jumps
Up	Rises through each cycle, then jumps back
Down	Falls through each cycle, then jumps back
Rand	Chooses a new target once per cycle and ramps toward it, rather than jumping between unrelated values every sample

LFO Phase offsets the point in the modulation cycle.

Try it

For clear echoes, set Depth to zero, moderate Feedback and the filter fairly open. For gentle chorus-like motion, use a short Time, Sine or Tri, slow LFO Freq and a small Depth. For more radical pitch sweeps, raise Depth or use Up/Down. Freeze a recognizable fragment, then release Freeze to let new material enter again.

Pitch Delay



A delay with a moving-read-head pitch shifter.

Pitchshift and Octave combine to set playback pitch;

Loop Depth sets the pitch-shifting window as a portion of the delay time.

Encoder	Control
Blue	Time: 1 ms–1 second
Green	Feedback: 0–200%
White	Pitchshift: equal-tempered steps and ratios from unison to one octave up
Orange	Mix

White push switches the white encoder to Octave (-3 to +3 octaves),

Orange push switches the orange encoder to Loop Depth (10–100%), shown below Mix. Push again to return to Mix. Shorter windows give a different, more granular pitch-shifting texture.

Pitchshift, Octave and Loop Depth

Pitch Delay follows the early moving/rotary-head approach described by Tom Erbe, related to devices such as the Tempophon.

Two read heads move through the delay and are crossfaded to soften the points where a head wraps back. Its changing windows and characteristic artifacts are part of the effect; extreme shifts can sound rough or distorted.

Time sets the underlying delay and contributes to the available pitch-shifting window.

Longer times generally give the moving-head process more material to work with. Shorter times can produce a tighter, more metallic or fragmented sound. The OP-1 range is 1 ms–1 second.

Pitchshift includes both equal-tempered semitone positions and ratio positions between unison (1:1) and one octave up (2:1).

The screen shows semitone choices as +n st and ratio choices as fractions. For example, +7 st is an equal-tempered fifth, while 3/2 is an exact frequency ratio. They are close but not identical.

Octave adds or subtracts up to three octaves from that base shift. At Pitchshift 0 st, Octave -1 halves the playback pitch; Octave +1 doubles it. At large shifts, expect stronger artifacts and a less recognizable source.

Loop Depth selects how much of the delay line is used for the shifting window, from 10–100% of Time. It changes the window duration rather than the dry/wet balance.

For example, Time 400 ms and Loop Depth 25% give a nominal 100 ms window. Shorter windows can make the pitch-shifting texture more obvious; larger windows use longer fragments.

Feedback returns the shifted output to the delay, so the shift can accumulate through successive repeats. A fifth-up setting with moderate Feedback can produce a rising succession of fifths.

An octave-down setting can create progressively deeper repeats. Above 100%, feedback can build strongly and saturation becomes more prominent.

Try it

Start at 0 st, Octave 0 and moderate Feedback. Select +7 st or 3/2 to hear rising fifths in the repeats. Then try Octave -1 for falling layers. Change Loop Depth while keeping Pitchshift fixed to hear the difference between shorter and longer windows.

Differences from the original SoundHack VSTs

Original feature or range	OP-1 adaptation
+delay Time up to 5 seconds; +bubbler up to 10 seconds	All three base Time controls stop at 1 second
+bubbler Time begins at 20 ms	Bubbler begins at 30 ms
+bubbler Density 0–200%, averaging two grains at 100%	Density 0–100%, with an engine limited to eight simultaneous grains
+bubbler Size 0–50%	Size 2–50%, with a minimum grain length
Original +bubbler Time Var formula	The port uses a bounded random spread and valid-history limits
Multiple pitch buttons and octave choices, 12tet/just switch	One Bubbler base Pitch choice, key mode and random octave probability; no chord-button or tuning-system selector
Stereo LFO Phase spread	Delayed's wet engine is mono; Phase offsets its modulation cycle
Freeze retains modulation and filtering	Delayed directly recirculates its captured frozen loop
Auto Time, feedback polarity, MIDI Time/LFO Lock, selectable one/two-head mode in +delay	These controls are not exposed by Delayed

These devices preserve the musical ideas of the original trio while adapting controls, numeric processing and memory use for the original OP-1. The original VST manual is useful background, but its ranges and unexposed controls should not be assumed to apply here.

Credits

Based on **SoundHack Delay Trio by Tom Erbe**, using the **Opus framework**. Explanations are adapted and paraphrased from Tom Erbe's *SoundHack Delay Trio* manual (2008), especially +delay on pages 2–3, +pitchdelay on page 4 and +bubbler on pages 5–6. Original concepts and algorithms are credited to Tom Erbe / SoundHack. OP-1 ports by **ESC_CODE**. See Credits for links and licence notices.

For further background, the original manual cites Curtis Roads, *Microsound* (MIT Press, 2001), and Dennis Gabor, "Acoustical quanta and the theory of hearing" (*Nature*, 1947).

Full credits

Tom Erbe / SoundHack — original author of the SoundHack Delay Trio: **+bubbler, +delay and +pitchdelay**. These devices are ports and adaptations of his work. Thank you for making the original source available. SoundHack

LizardMan / LizardMan245 and Opus contributors — the Opus framework, original OP-1 integration, SDK, plugin kit and examples that make these devices possible. Opus

ESC_CODE / Volker Junginger — OP-1 ports, control mappings, interface adaptation and optimization. Artwork was developed from approved generated concepts and adapted for the OP-1 renderer.

SquareWave Studio — OP-1 Design Kit and stock-inspired screen design guidance. OP-1 Design Kit

The ports include GPL-3.0-or-later code; see LICENSE and individual source notices. The original SoundHack code remains covered by its original BSD-style licence; see SoundHack-LICENSE.txt. Retain both notices when redistributing. Opus is credited under its own source licence; its build kit is obtained separately.

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